Pavan Satya Krishna Venigalla

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Education

Rutgers University — Camden, NJ
Master of Science in Data Science; GPA: 4.0/4.0 — *Aug 2023 – May 2025*

National Institute of Technology Tiruchirappalli — India
Bachelor of Engineering in Instrumentation and Control Engineering — *Jul 2019 – May 2023*

Experience

Los Angeles County Department of Health Services — Los Angeles, CA
Informatics Data Analyst — *Nov 2025 – Present*

- Led end-to-end migration of clinical and operational data from Oracle Millennium HealtheIntent to Azure cloud, developing ETL/ELT pipelines using PySpark, T-SQL, and Azure Synapse aligned with Medallion Architecture (Bronze/Silver/Gold) patterns.
- Designed SCD-style historical tracking and Delta Lake optimizations (liquid clustering, Z-order indexing) to improve query performance; built automated CI/CD deployment pipelines using Azure DevOps YAML across dev/staging/production environments.
- Developed data quality validation and UAT frameworks for pipeline outputs; conducted root cause analysis on quality measure drops across CBP and CMS metrics using HEDIS, CPT, and LOINC codes.

I Net Software Technologies Inc — TX, USA
Data Engineer Intern — *Jun 2025 – Oct 2025*

- Designed and optimized ETL pipelines and data models using SQL, Python, and Excel to enable accurate reporting, dashboarding, and ad-hoc analysis for client stakeholders.
- Built Power BI dashboards for performance tracking and operational KPIs; automated data preparation workflows to reduce manual effort and improve turnaround time.

Rutgers University — NJ, USA
Research Data Analyst — *Jan 2024 – May 2025*

- Analyzed large biological datasets using SQL, Python, and Excel; performed statistical analysis, hypothesis testing, and regression modeling to support research conclusions.
- Built data visualizations and analytical summaries; conducted trend analysis and predictive modeling for lab equipment utilization forecasting.

Piramal Pharma — Bangalore, India
Data Analyst – Healthcare Analytics — *Jun 2022 – Jul 2023*

- Developed scalable data pipelines using GCP BigQuery and Python to process and standardize healthcare operational, patient, and treatment datasets from multiple sources.
- Built dimensional data models and data marts supporting analytics, regulatory reporting, and patient outcome monitoring.
- Created interactive Power BI dashboards to visualize patient treatment trends, therapy adherence metrics, and operational KPIs for business stakeholders.

Projects

NutriWeb | ***Python, NLP, LLM APIs, scikit-learn, pandas***

- Built a dietary recommendation system using NLP-based food entity extraction and LLM APIs to score, rank, and match food options against structured nutrition datasets.
- Designed matching algorithms combining text similarity scoring and nutritional profiling to align recommendations with user dietary preferences and health goals.

Technical Skills

Languages: Python, PySpark, T-SQL, PL-SQL, KQL, R, SQL, JavaScript
Databases: Azure Synapse Analytics, GCP BigQuery, SQL Server, MySQL, DynamoDB
Cloud & DevOps: Azure (ADLS Gen2, Synapse, Fabric, DevOps), GCP, AWS, CI/CD (YAML Pipelines)
Data Architecture: Medallion Architecture, Delta Lake, Star Schema, Dimensional Modeling
BI & Visualization: Power BI (DP-700 Certified), Tableau
AI Tools: GitHub Copilot, Amazon Bedrock, Claude

Certifications

Microsoft DP-700 Fabric Analytics Engineer | AWS Certified AI Practitioner | Google Data Analytics Professional